Forest Resources



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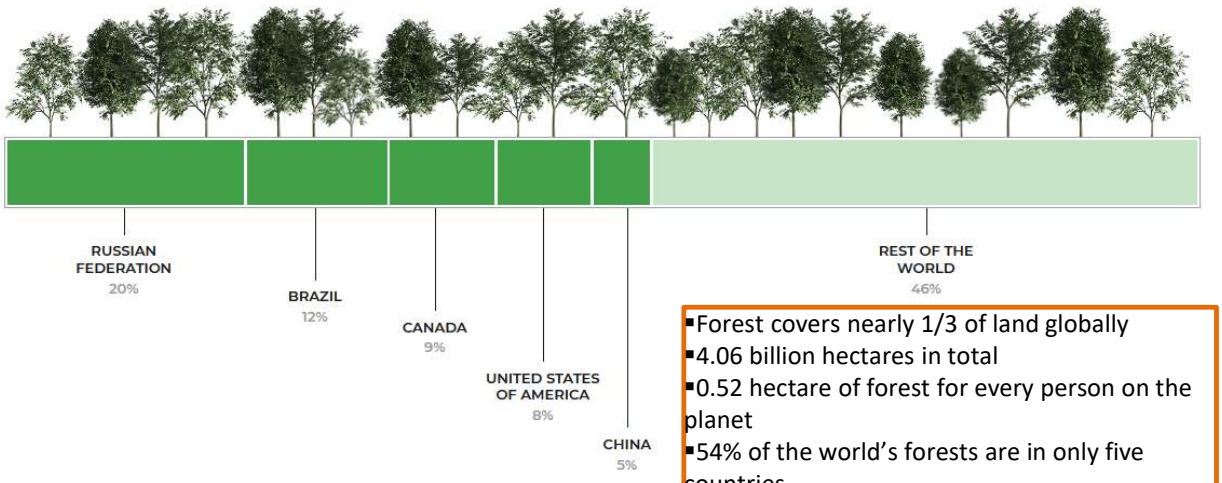
Learning Objectives

After taking this class, the students will be able to:

- *Acknowledge the condition of forest health*
- *Understand why forests are important to us*
- *Be familiar with policy instruments for forest conservation*
- *Understand the difficulties to deal with the social ecological system*

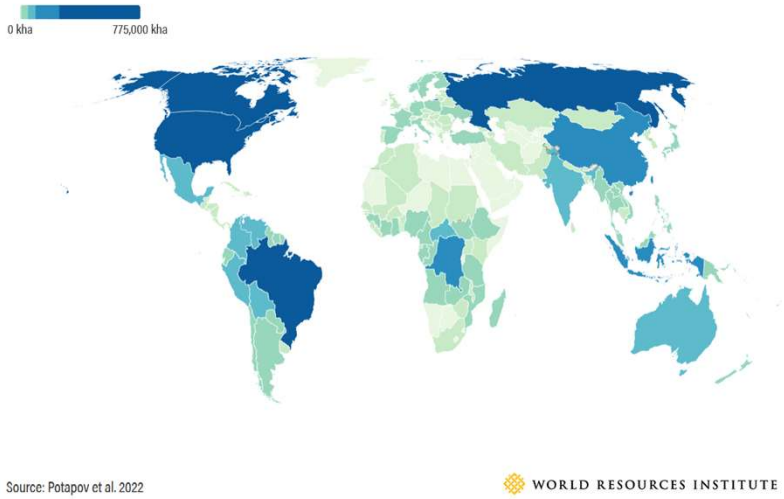
Forest Condition and Health

Global Forest Condition



- Forest covers nearly 1/3 of land globally
- 4.06 billion hectares in total
- 0.52 hectare of forest for every person on the planet
- 54% of the world's forests are in only five countries

Global Forest Condition



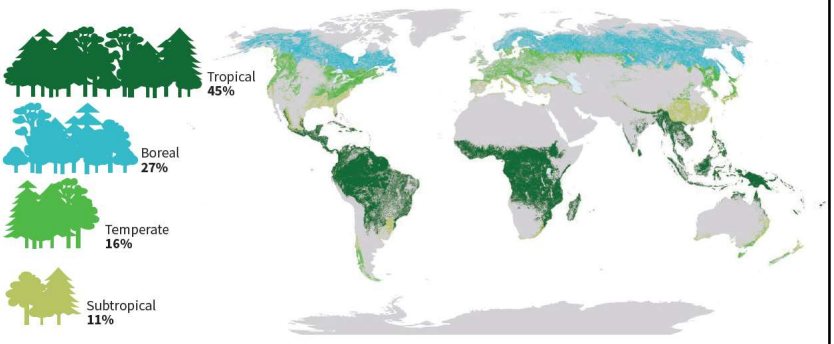
Source: Potapov et al. 2022

WORLD RESOURCES INSTITUTE

According to your perception of forests worldwide, which climatic region do you think contains the most forest?

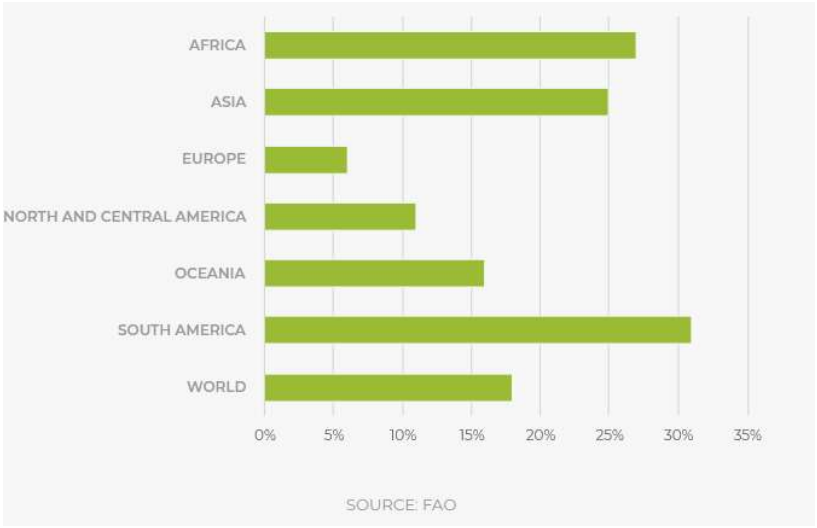
- Boreal
- Temperate
- Tropical
- Subtropical

Proportion and distribution of global forest area by climatic domain, 2020

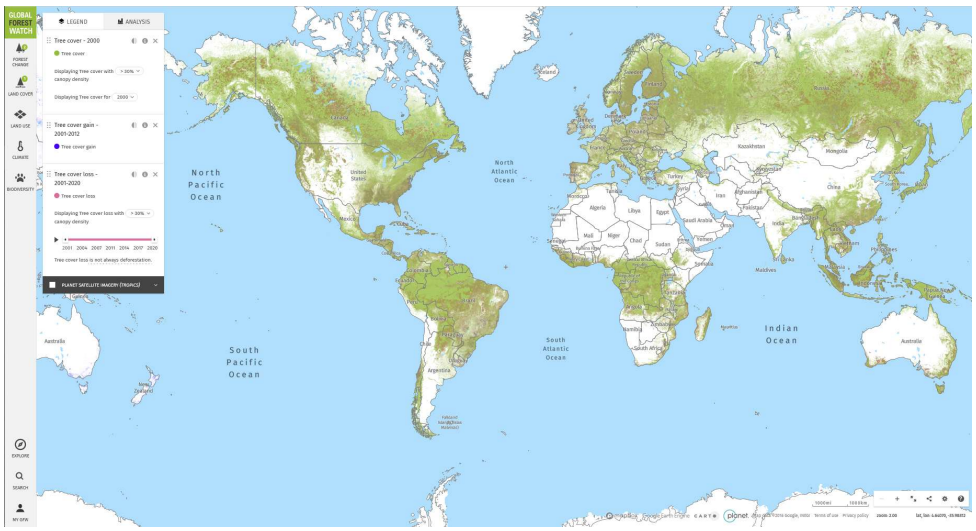


According to your perception of the world's forests, which region do you think has the highest share of forest in protected areas?

- Africa
- Asia
- Europe
- North and Central America
- Oceania
- South America



Global Forest Watch



How are forests different from tree cover?

- Tree cover is a convenient metric for monitoring forest change because it is easily measurable from space using freely available, medium-resolution satellite imagery.
- The **existence of tree cover does not always make a forest**, tree cover loss does not always imply forest loss or deforestation, and tree cover gain does not always imply forest gain or restoration.
- Most definitions of forest involve a combination of tree cover and land use. The latter is much more difficult, if not impossible in some cases, to monitor using satellite imagery.



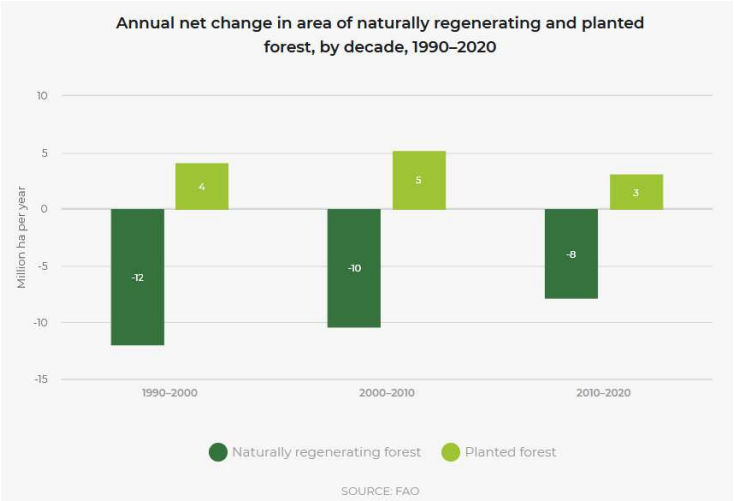
Types of Forests

- Naturally Regenerating Forest (89%)
  - Primary forest: A forest of native tree species, where there are *no visible indications of human activities*, and the ecological processes are not significantly disturbed.
  - Primary forests account for roughly 50 percent of all forests in the tropics (1.03 billion hectares).
  - Brazil, the Democratic Republic of the Congo and Indonesia have the most primary forest in absolute terms.
  - 41 percent of global forests are managed natural forests, while 48 percent are unmanaged natural forests.
- Planted Forest (11%)
  - A forest that is *intensively managed*, is composed of one or two species of even ages with regular spacing.



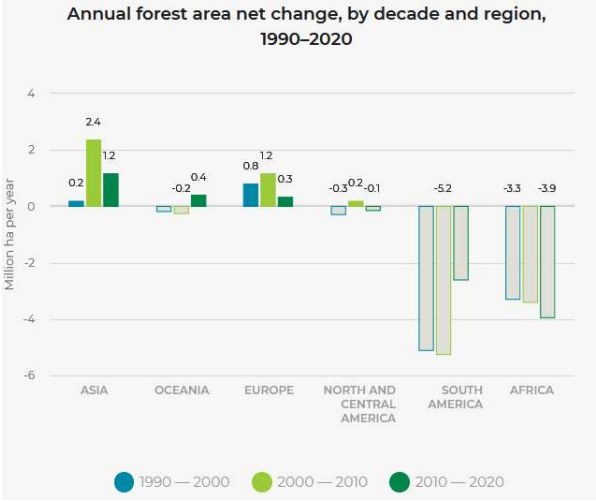


Forest Losses and Gains



- The area of primary forest has decreased by 81 million ha since 1990.
- The loss of naturally regenerating forest has slowed down.
- The loss of naturally regenerating forest is more than the gain of planted forest.

Forest Losses and Gains



- Asia had the highest net gain of forest area in 2010–2020.
- The global forest area declined by about 178 million ha in the 30 years from 1990 to 2020.
- The world's forest area is decreasing, but the rate of loss has slowed.
- The rate of net forest loss declined from 7.8 million ha per year in the decade 1990–2000 to 5.2 million ha in 2000–2010 and 4.7 million ha per year in 2010–2020.

Forest Losses and Gains

Top ten countries for average annual net loss of forest area, 2010-2020

Ranking	Country	%
1	Brazil	-0.30
2	Democratic Republic of the Congo	-0.83
3	Indonesia	-0.78
4	Angola	-0.80
5	United Republic of Tanzania	-0.88
6	Paraguay	-1.93
7	Myanmar	-0.96
8	Cambodia	-2.68
9	Bolivia (Plurinational State of)	-0.43
10	Mozambique	-0.59

Note: The rate of change (%) is calculated as the compound annual change rate.

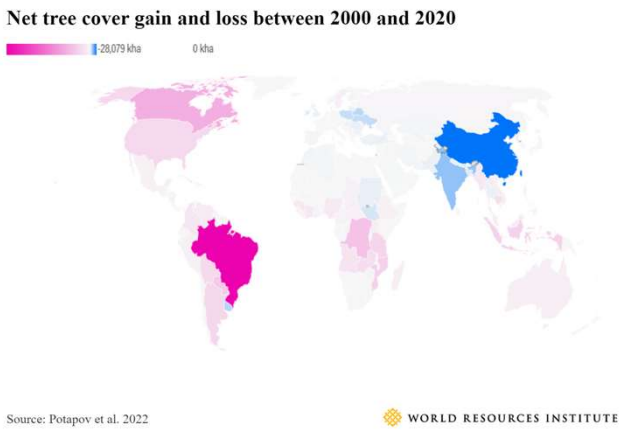
Top ten countries for average annual net gain in forest area, 2010-2020

Ranking	Country	Annual net change	
		1 000 ha/yr	%
1	China	1 937	0.93
2	Australia	446	0.34
3	India	266	0.38
4	Chile	149	0.85
5	Viet Nam	126	0.90
6	Turkey	114	0.53
7	United States of America	108	0.03
8	France	83	0.50
9	Italy	54	0.58
10	Romania	41	0.62

Note: The rate of change (%) is calculated as the compound annual change rate.

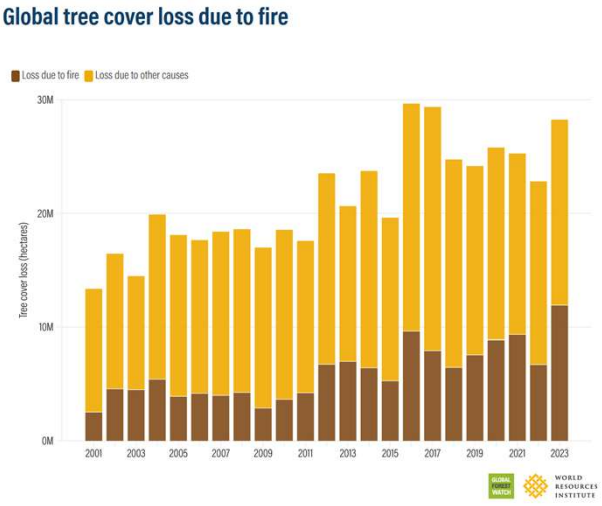
Forest Losses and Gains

- The world has lost 488 Mha of tree cover since the turn of the century, equivalent to about 12 percent of global tree cover in 2000.
- Tree cover loss has been rising in recent history, from 13.4 Mha of tree cover loss in 2001 to 28.3 Mha in 2023.
- Roughly 27% of tree cover loss since 2000 was likely to be deforestation, characterized by permanent loss of forest resulting in new, nonforest land uses such as agriculture and human settlements.
- Though only half (239 Mha) of global tree cover loss this century occurred in the tropical ecozone, the tropics accounted for nearly 97 percent of all global deforestation.



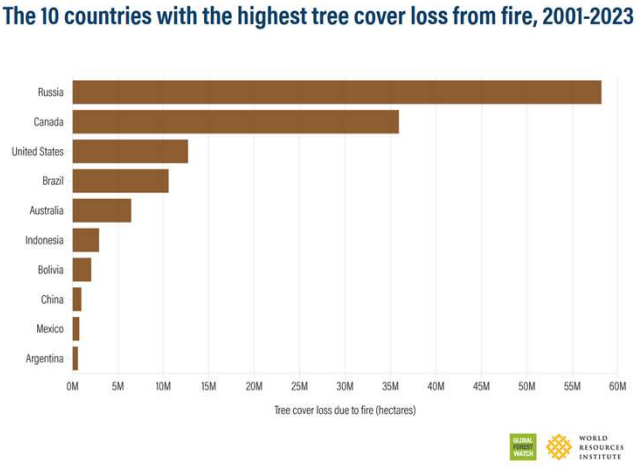
Global Tree Cover Loss due to Fire

- Worldwide, 138 Mha of tree cover loss was caused by fire between 2001 and 2023, representing over a quarter of all tree cover loss during this period.
- Forest loss due to fire has increased since 2001, driven in part by climate change as increasingly warmer and drier conditions increase fire activity in some parts of the world.



Global Tree Cover Loss due to Fire

- Just five countries — Australia, Brazil, Canada, Russia and the United States — accounted for nearly 90 percent of all tree cover loss due to fire globally.
- Tree cover loss related to wildfire is not necessarily permanent.
- In some fire-adapted forest ecosystems, it can be beneficial for long term forest health, while in other contexts it can cause forest degradation.



Fire and Risk Alerts to Forests





- There were **58,838 VIIRS** fire alerts reported in **China** between **24th of February 2025** and **24th of March 2025**, of which **1.9%** were high confidence alerts.
- Between **24th of February 2025** and **24th of March 2025 China** experienced a total of **1,117 VIIRS Alerts** fire alerts, considering **high confidence** alerts only.

Reflective Question:

Is tree planting always good?

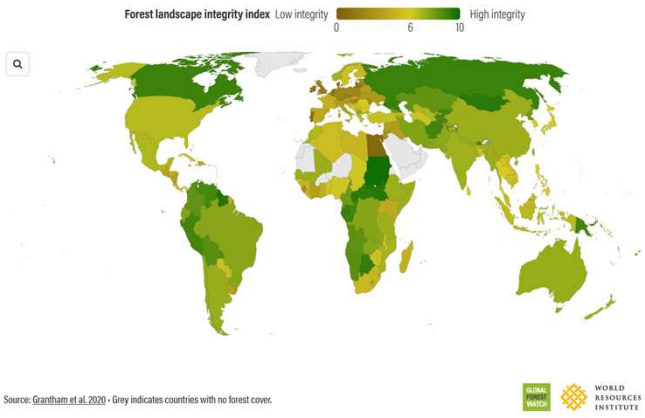
Tree Planting and Climate Change



Forest Condition and Health

- **Forest condition** considers the quality of standing forests.
- **Forest landscape integrity** estimates forest condition based on the degree of human modification by integrating data on observed and inferred human pressures and changes to forest connectivity.
- As of 2019, 40.5 percent of forests globally had **high integrity**.
- Approximately a quarter had **low integrity** (25.6 percent), and the remaining 33.9 percent had **medium integrity**.

Average forest landscape integrity score by country

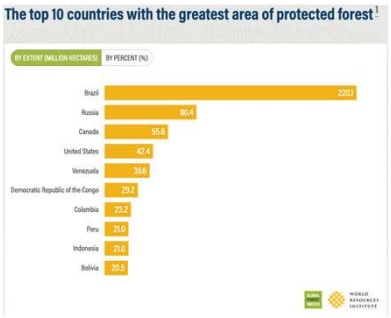


How do we manage forest resources?

Government in Forest Management

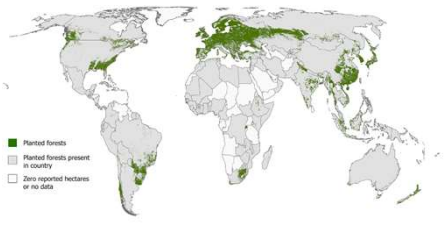
Legally Protected Forests

- Approximately 21 percent of global forest area is currently under some form of legal protection.
- Countries with the largest amount of forest overall—such as Brazil and Russia—also tend to have the largest amount of forest under protection.

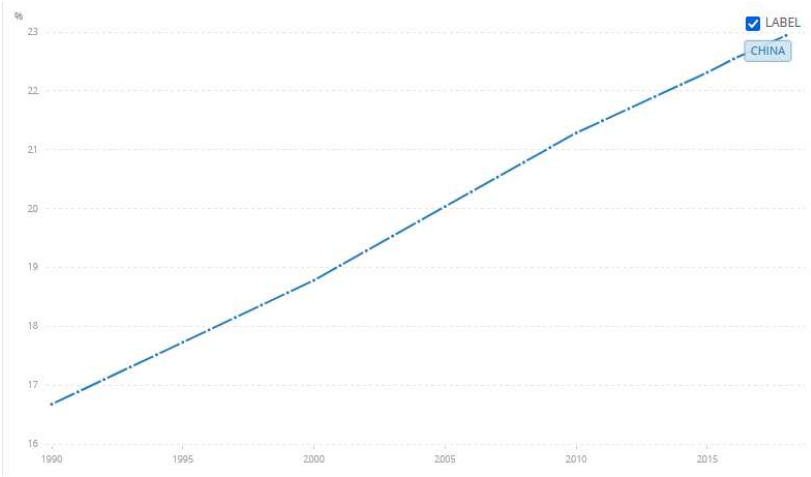


Production Forests

- An estimated 1.15 Bha of forest are designated primarily for production, equivalent to roughly 30 percent of global forest area.
- Production forests appear extensively in the temperate and boreal forests of North America, Europe, Russia, China, southern Brazil, Chile, South Africa, and Australia.



Achievements of Forest Management in China



(World Bank)

Policy Instruments for Improving Tree Cover

- Six Major Projects on Forest Protection (since 1999)
  - Natural Forest Protection Program （天然林资源保护工程）
  - Conversion of Cropland to Forest Program （退耕还林工程）
  - Sandification Control Program in the Vicinity of Beijing and Tianjin （京津风沙源治理工程）
  - Key Shelterbelts Development Programs in the Middle and Lower Reaches of Yangtze River （三北和长江中下游地区等重点防护林建设工程）
  - Wildlife and Nature Reserve Management Program （野生动植物保护及自然资源保护区建设工程）
  - Fast-Growing and High-Yielding Timber Base Construction Program in Key Areas （重点地区速生生产用材林基地建设工程）
- Compulsory Targets in the Five-Year Plan

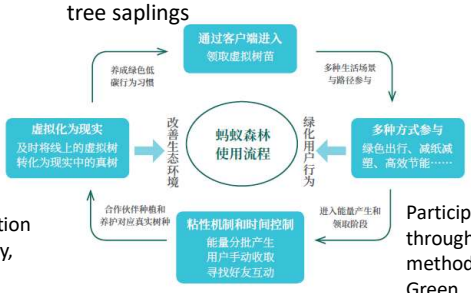
FYP	Tree Cover	Forest stock
12th FYP (2011 - 2015)	21.66%	> 14.3 Bm^3
13th FYP (2016 - 2020)	23.04%	>15.24 Bm^3
14th FYP (2021 - 2025)	24.1%	>19 Bm^3

Non-Governmental Actors for promoting forest conservation

- Have you play with Ant Forest (蚂蚁森林), Xiaodu Farm (小度农庄), or “Panda Protector”(熊猫守护者)?
- According to the data of 2019, there are more than 500 million users of Ant Forest. More than 122 million trees are planted.
- You can make some exploration on these programs.

Case of Ant Forests

Accessing through the client-side application, users can claim virtual tree saplings

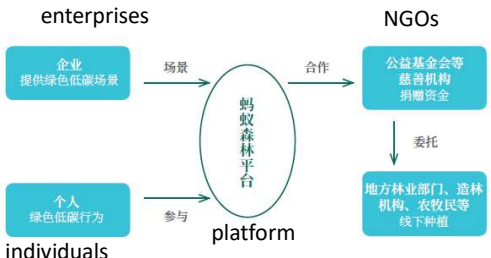


Virtualization into reality, promptly transforming virtual trees online into real trees in the physical world

Sticky mechanism and time control

Participation through various methods: Green transportation, paper and plastic reduction

The user workflow for Ant Forest



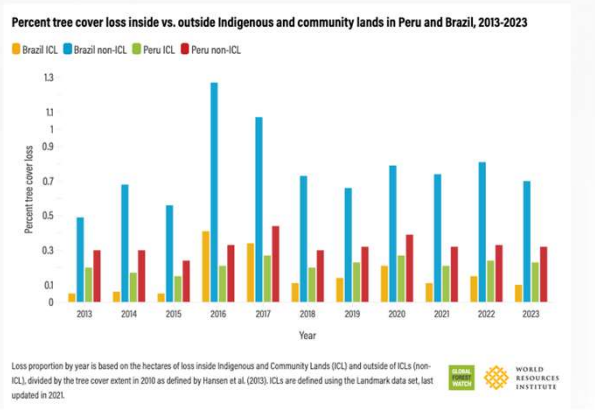
Local forest management bureau, Afforestation organization farmers,

The operational process of the Ant Forest platform

Non-Governmental Actors for promoting forest conservation

- It is estimated that between one-fifth and one-third of the world's remaining intact forest landscapes are on Indigenous land.
- At least 36% of the global Key Biodiversity Areas (KBAs) lie within Indigenous Peoples and local community lands.
- Indigenous and community lands in Brazil, Peru, Mexico and Colombia sequestered more than double the amount of carbon per hectare than other areas.
- Between 2013 and 2023, the percentage of tree cover loss outside of Indigenous and community lands in Brazil and Peru was higher than within.

Tree cover loss is lower inside Indigenous and community lands in Peru and Brazil than outside [8]



Ecosystem Services for Poverty Alleviation

A comparison between counties with key ecological functional zones and nationally designated poverty-stricken counties in Southwest and Northwest China

表 1　西北地区、西南地区重点生态功能区所在县和国家级贫困县的比对分析

		重点生态功能区所在县数量 Number of Counties with Key Ecological Functional Zones	国家级贫困县数量 Number of Nationally Designated Poverty-Stricken Counties	重叠数量 overlap	重点生态功能区所在县中国家级贫困县的比例(%) The ratio of counties with key ecological functional zones to nationally designated poverty-stricken counties.	国家级贫困县中重点生态功能区所在县的比例(%) The proportion of nationally designated poverty-stricken counties among counties with key ecological functional zones
Chongqing	重庆	9	14	9	100	64
Sichuan	四川	56	66	51	91	77
Guizhou	贵州	46	66	41	89	62
Yunnan	云南	46	88	42	91	48
Tibet	西藏	36	74	35	97	47
Shaanxi	陕西	43	56	36	84	64
Gansu	甘肃	48	58	36	75	62
Qinghai	青海	41	42	41	100	98
Ningxia	宁夏	12	8	8	67	100
Xinjiang	新疆	49	32	25	51	78
Total	合计	386	504	324	84	64

数据来源：财政部公布的 2017 年重点生态功能区转移支付范围县域名单，国务院扶贫开发领导小组办公室 2014 年公布的全国 832 个贫困县名单。

Discussion:

How to balance economic development and forest conservation?

Payments for Ecosystem Services (PES)

- Payments for Ecosystem Services
  - a variety of arrangements through which the beneficiaries of environmental services, from watershed protection and forest conservation to carbon sequestration and landscape beauty, reward those whose lands provide these services with subsidies or market payments.

Another View on Tree Plantation



References

- Food and Agriculture Organization of the United Nations, 2021, Global Forest Resources Assessment 2020
- World Resources Institute, 2023, Global Forest Review: Reporting on the status of the world's forests
- 胡振通、王亚华，2021，中国生态扶贫的理论创新和实现机制，清华大学学报（哲学社会科学版）36， 1。
- 生态环境部环境与经济政策研究中心课题组，2019，互联网平台背景下公众低碳生活方式研究报告。